

# Frozen grouped factor model (OOS 2023+)

Directed factor-mimicking portfolio catalogue

## Abstract

This catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by reported raw Sharpe, highest first. Direction is frozen upstream; this report performs no return-based sign selection.

## Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes the reported Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only. Return Uncertainty uses paired circular stationary-bootstrap draws of the common monthly panel. It measures sampling and path uncertainty, not out-of-sample generalisation.

## Construction Configuration

| Setting            | Effective value                                                           |
|--------------------|---------------------------------------------------------------------------|
| Factors            | 5                                                                         |
| Universe           | shared security screen; refreshed monthly                                 |
| Formation filters  | price $\geq$ None; 21d dollar volume $\geq$ none                          |
| Market-cap screen  | top 90.0% of names monthly                                                |
| Signal preparation | within-month winsorize, normalize, then demean                            |
| Composite          | mean available normalized components; factor-specific coverage floor      |
| Construction       | one blanket profile                                                       |
| Rebalance / return | month-end / next calendar month simple return                             |
| Costs              | effective configured costs; default 0 bps                                 |
| Minimum sample     | 20 stocks/month; 24 common months; 90% factor coverage                    |
| Comparison window  | 2023-01-01 to 2026-05-01; factor- and control-complete                    |
| Winsor settings    | 1%-99%                                                                    |
| Controls           | MKT, SMB, VAL, MOM, BAB, QMJ                                              |
| Return uncertainty | paired stationary bootstrap; 2000 draws; expected block 12 months; seed 0 |

## Returns PCA

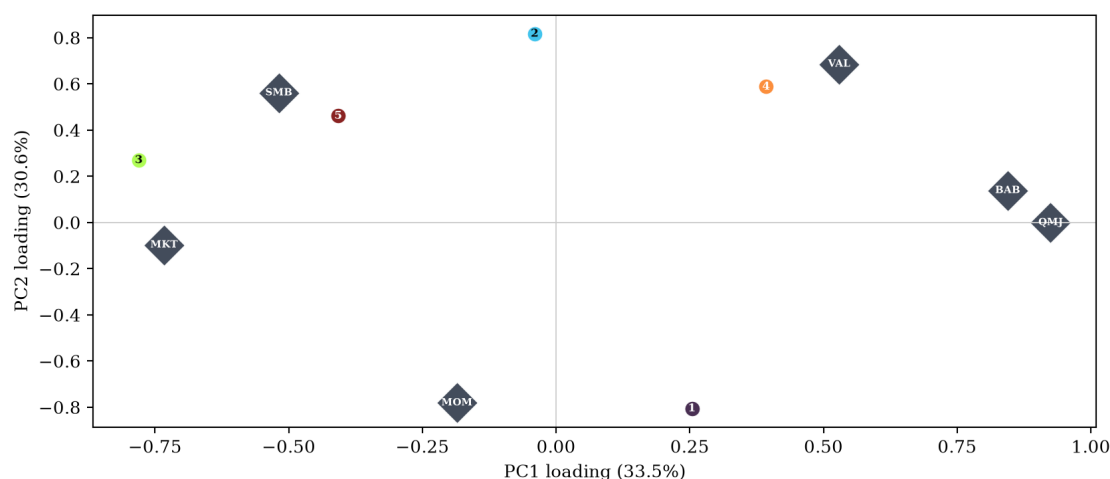


Figure: catalogue factors are numbered by reported-Sharpe index; attribution controls are named;  $N=41$  common months.

# Index

| Idx | Factor   | Dir. | N  |
|-----|----------|------|----|
| 1   | Factor_3 | +1   | 41 |
| 2   | Factor_2 | +1   | 41 |
| 3   | Factor_1 | +1   | 41 |
| 4   | Factor_4 | +1   | 41 |
| 5   | Factor_5 | +1   | 41 |

# Returns

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD  | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|---------|----------|-------------|------------|
| Factor_3 | 8.48%     | 9.33%     | 0.91   | 2.28  | -5.93%  | 0.683    | 5.34%       | 9.33%      |
| Factor_2 | 2.40%     | 8.48%     | 0.28   | 0.75  | -8.05%  | 0.597    | 0.50%       | 8.48%      |
| Factor_1 | -1.05%    | 10.51%    | -0.10  | -0.19 | -25.53% | 0.715    | 0.26%       | 10.51%     |
| Factor_4 | -1.77%    | 3.83%     | -0.46  | -0.94 | -9.51%  | 0.480    | -0.35%      | 3.83%      |
| Factor_5 | -3.07%    | 6.47%     | -0.47  | -1.01 | -17.31% | 0.261    | -3.10%      | 6.47%      |

# Return Uncertainty

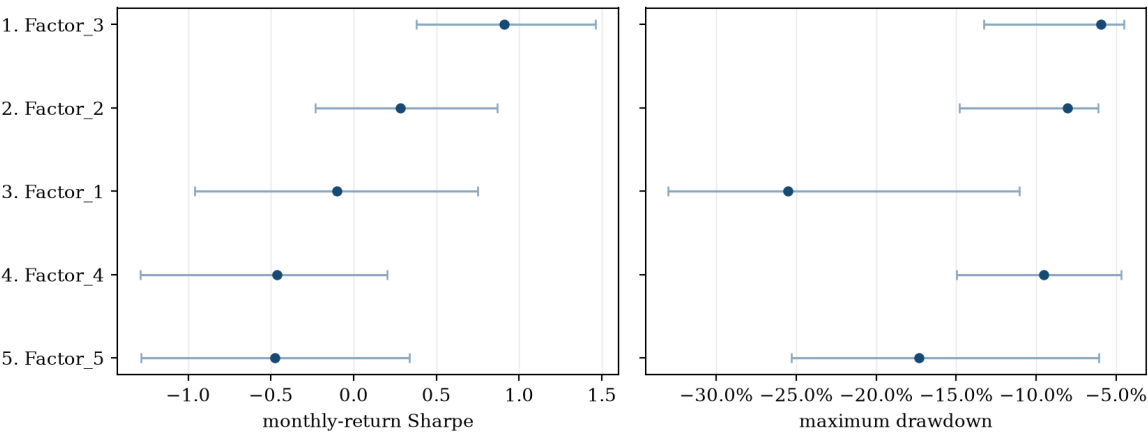


Figure: point estimates and paired stationary-bootstrap 95% intervals.

| Idx | Factor   | Ann. ret. [95%]     | Sharpe [95%]        | Pr(S<=0) | Skew [95%]          | Max DD [95%]            | ES 5% [95%]          |
|-----|----------|---------------------|---------------------|----------|---------------------|-------------------------|----------------------|
| 1   | Factor_3 | 8.5% [3.7%, 13.4%]  | 0.91 [0.38, 1.46]   | 0.1%     | -0.06 [-0.66, 0.72] | -5.9% [-13.3%, -4.5%]   | -4.3% [-6.8%, -2.3%] |
| 2   | Factor_2 | 2.4% [-2.1%, 7.0%]  | 0.28 [-0.23, 0.87]  | 14.3%    | -0.24 [-0.69, 0.32] | -8.1% [-14.8%, -6.1%]   | -4.3% [-6.1%, -3.1%] |
| 3   | Factor_1 | -1.1% [-9.5%, 7.7%] | -0.10 [-0.96, 0.75] | 61.3%    | -0.08 [-0.54, 0.39] | -25.5% [-33.0%, -11.1%] | -5.4% [-6.0%, -4.3%] |
| 4   | Factor_4 | -1.8% [-4.4%, 0.8%] | -0.46 [-1.29, 0.20] | 91.8%    | 0.34 [-0.09, 0.86]  | -9.5% [-15.0%, -4.7%]   | -2.0% [-2.5%, -1.6%] |
| 5   | Factor_5 | -3.1% [-7.8%, 2.3%] | -0.47 [-1.28, 0.34] | 87.9%    | 0.05 [-0.39, 0.51]  | -17.3% [-25.3%, -6.1%]  | -3.8% [-4.8%, -2.3%] |

# Risk Loadings

| Factor    | MKT              | SMB              | VAL              | MOM              | BAB              | QMJ              | Alpha             | R2    |
|-----------|------------------|------------------|------------------|------------------|------------------|------------------|-------------------|-------|
| Factor__3 | -0.16<br>(-1.94) | -0.17<br>(-1.60) | -0.18<br>(-2.07) | 0.42<br>(7.11)   | 0.52<br>(2.75)   | -0.18<br>(-1.05) | 5.39%<br>(1.74)   | 0.683 |
| Factor__2 | 0.32<br>(2.88)   | 0.07<br>(0.56)   | 0.47<br>(3.41)   | -0.15<br>(-1.86) | -0.16<br>(-0.61) | 0.09<br>(0.50)   | 0.18%<br>(0.07)   | 0.597 |
| Factor__1 | -0.21<br>(-1.59) | 0.37<br>(7.14)   | 0.13<br>(1.79)   | 0.07<br>(1.15)   | -0.43<br>(-1.95) | -0.49<br>(-2.63) | -0.87%<br>(-0.21) | 0.718 |
| Factor__4 | 0.08<br>(1.39)   | 0.15<br>(3.60)   | 0.10<br>(1.87)   | -0.02<br>(-0.64) | 0.02<br>(0.38)   | 0.18<br>(2.59)   | 0.02%<br>(0.01)   | 0.484 |
| Factor__5 | 0.16<br>(1.82)   | 0.07<br>(0.60)   | 0.02<br>(0.14)   | -0.17<br>(-3.95) | -0.12<br>(-0.53) | -0.01<br>(-0.09) | -4.87%<br>(-1.38) | 0.284 |

# Return Contributions

| Factor    | MKT    | SMB    | VAL    | MOM    | BAB    | QMJ    | Alpha  | Total  |
|-----------|--------|--------|--------|--------|--------|--------|--------|--------|
| Factor__3 | -3.54% | 0.85%  | 0.92%  | 4.22%  | -1.33% | 1.97%  | 5.39%  | 8.48%  |
| Factor__2 | 7.05%  | -0.36% | -2.37% | -1.53% | 0.40%  | -0.97% | 0.18%  | 2.40%  |
| Factor__1 | -4.75% | -1.89% | -0.68% | 0.69%  | 1.10%  | 5.36%  | -0.87% | -1.05% |
| Factor__4 | 1.72%  | -0.77% | -0.49% | -0.23% | -0.06% | -1.96% | 0.02%  | -1.77% |
| Factor__5 | 3.52%  | -0.34% | -0.08% | -1.73% | 0.30%  | 0.12%  | -4.87% | -3.07% |

# 1. Factor\_3

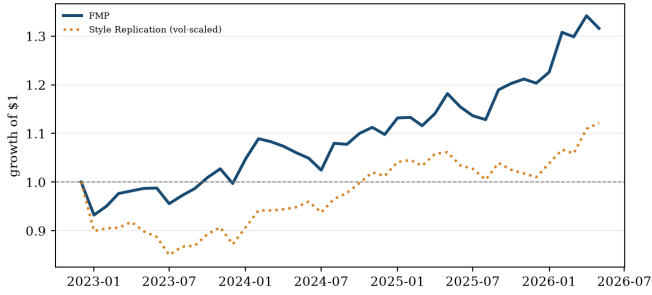


Figure: Common-window cumulative growth of \$1.

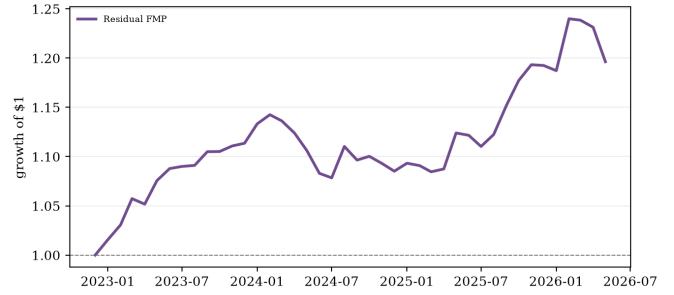


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|--------|----------|-------------|------------|
| Factor_3 | 8.48%     | 9.33%     | 0.91   | 2.28  | -5.93% | 0.683    | 5.34%       | 9.33%      |

| Factor   | MKT              | SMB              | VAL              | MOM            | BAB            | QMJ              | Alpha           | R2    |
|----------|------------------|------------------|------------------|----------------|----------------|------------------|-----------------|-------|
| Factor_3 | -0.16<br>(-1.94) | -0.17<br>(-1.60) | -0.18<br>(-2.07) | 0.42<br>(7.11) | 0.52<br>(2.75) | -0.18<br>(-1.05) | 5.39%<br>(1.74) | 0.683 |

| Factor   | MKT    | SMB   | VAL   | MOM   | BAB    | QMJ   | Alpha | Total |
|----------|--------|-------|-------|-------|--------|-------|-------|-------|
| Factor_3 | -3.54% | 0.85% | 0.92% | 4.22% | -1.33% | 1.97% | 5.39% | 8.48% |

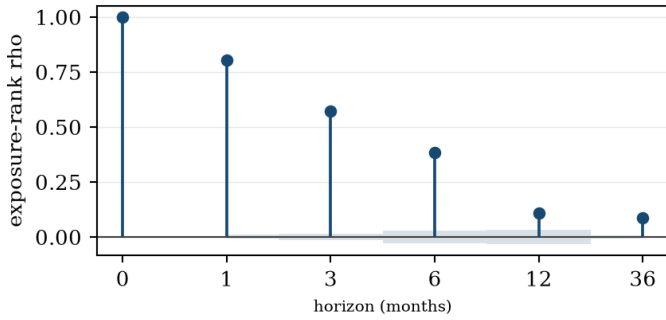


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

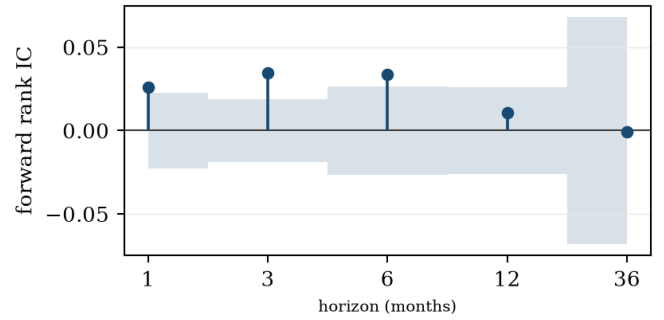


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------|
| Direction                     | +1                                                                                |
| Common months                 | 41                                                                                |
| Signals                       | Cluster_34(+1), Cluster_55(-1), Cluster_73(+1), Cluster_74(+1), Cluster_75(+1)    |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp |
| Components required           | 3 of 5                                                                            |
| Available / comparison months | 43 / 41                                                                           |
| Available window              | 2023-01-01 to 2026-07-01                                                          |
| Dropped: low breadth / common | 0 / 2                                                                             |
| Stocks, min / median / max    | 2106 / 2547 / 2811                                                                |
| Available-history Sharpe      | 0.61                                                                              |
| Bootstrap Sharpe              | 0.91 [0.38, 1.46]; Pr(S<=0)=0.1%                                                  |
| Bootstrap tails               | skew=-0.06 [-0.66, 0.72]; max DD=-5.9% [-13.3%, -4.5%]                            |

## 2. Factor\_2

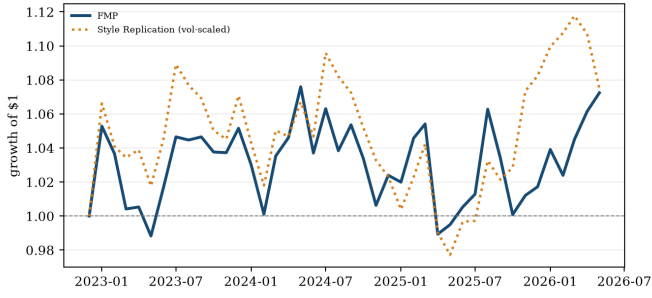


Figure: Common-window cumulative growth of \$1.

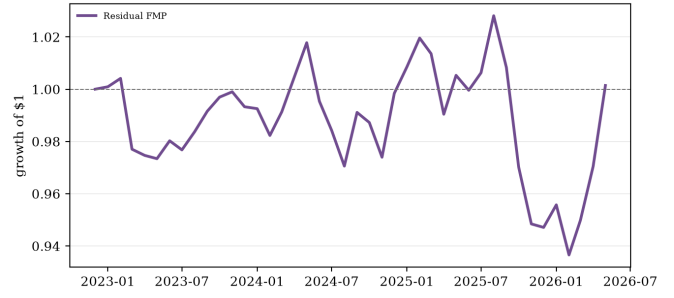


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|--------|----------|-------------|------------|
| Factor_2 | 2.40%     | 8.48%     | 0.28   | 0.75  | -8.05% | 0.597    | 0.50%       | 8.48%      |

| Factor   | MKT            | SMB            | VAL            | MOM              | BAB              | QMJ            | Alpha           | R2    |
|----------|----------------|----------------|----------------|------------------|------------------|----------------|-----------------|-------|
| Factor_2 | 0.32<br>(2.88) | 0.07<br>(0.56) | 0.47<br>(3.41) | -0.15<br>(-1.86) | -0.16<br>(-0.61) | 0.09<br>(0.50) | 0.18%<br>(0.07) | 0.597 |

| Factor   | MKT   | SMB    | VAL    | MOM    | BAB   | QMJ    | Alpha | Total |
|----------|-------|--------|--------|--------|-------|--------|-------|-------|
| Factor_2 | 7.05% | -0.36% | -2.37% | -1.53% | 0.40% | -0.97% | 0.18% | 2.40% |

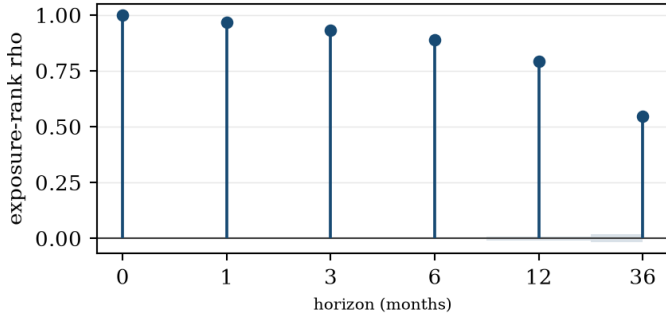


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

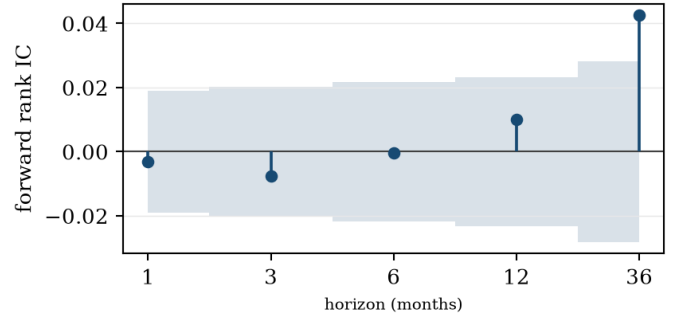


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction                     | +1                                                                                                                                                                                                                                                                                                                                                                           |
| Common months                 | 41                                                                                                                                                                                                                                                                                                                                                                           |
| Signals                       | Cluster_12(+1), Cluster_13(-1), Cluster_17(-1), Cluster_24(+1), Cluster_26(-1), Cluster_27(-1), Cluster_28(-1), Cluster_35(+1), Cluster_37(-1), Cluster_39(-1), Cluster_43(-1), Cluster_47(+1), Cluster_51(-1), Cluster_57(+1), Cluster_6(-1), Cluster_60(-1), Cluster_67(+1), Cluster_68(+1), Cluster_72(-1), Cluster_8(+1), Cluster_80(-1), Cluster_84(+1), Cluster_85(+1) |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp                                                                                                                                                                                                                                                                                            |
| Components required           | 14 of 23                                                                                                                                                                                                                                                                                                                                                                     |
| Available / comparison months | 43 / 41                                                                                                                                                                                                                                                                                                                                                                      |
| Available window              | 2023-01-01 to 2026-07-01                                                                                                                                                                                                                                                                                                                                                     |
| Dropped: low breadth / common | 0 / 2                                                                                                                                                                                                                                                                                                                                                                        |
| Stocks, min / median / max    | 2074 / 2431 / 2621                                                                                                                                                                                                                                                                                                                                                           |
| Available-history Sharpe      | 0.42                                                                                                                                                                                                                                                                                                                                                                         |
| Bootstrap Sharpe              | 0.28 [-0.23, 0.87]; Pr(S<=0)=14.3%                                                                                                                                                                                                                                                                                                                                           |
| Bootstrap tails               | skew=-0.24 [-0.69, 0.32]; max DD=-8.1% [-14.8%, -6.1%]                                                                                                                                                                                                                                                                                                                       |

### 3. Factor\_1

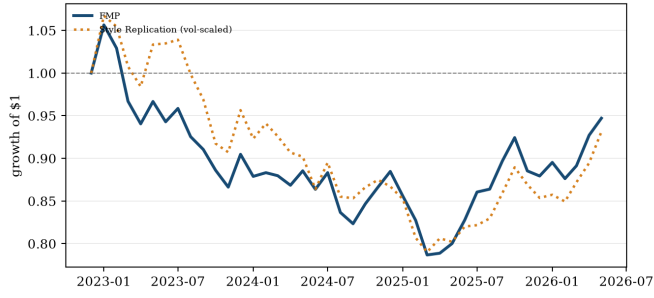


Figure: Common-window cumulative growth of \$1.

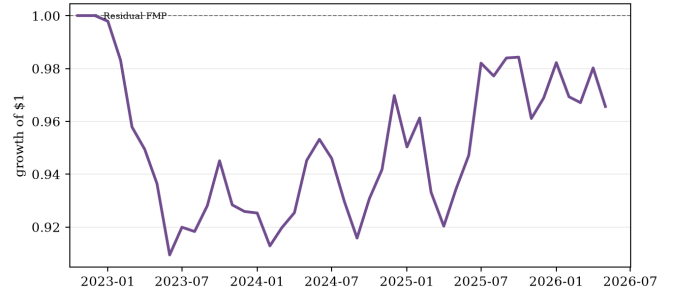


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD  | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|---------|----------|-------------|------------|
| Factor_1 | -1.05%    | 10.51%    | -0.10  | -0.19 | -25.53% | 0.715    | 0.26%       | 10.51%     |

| Factor   | MKT              | SMB            | VAL            | MOM            | BAB              | QMJ              | Alpha             | R2    |
|----------|------------------|----------------|----------------|----------------|------------------|------------------|-------------------|-------|
| Factor_1 | -0.21<br>(-1.59) | 0.37<br>(7.14) | 0.13<br>(1.79) | 0.07<br>(1.15) | -0.43<br>(-1.95) | -0.49<br>(-2.63) | -0.87%<br>(-0.21) | 0.718 |

| Factor   | MKT    | SMB    | VAL    | MOM   | BAB   | QMJ   | Alpha  | Total  |
|----------|--------|--------|--------|-------|-------|-------|--------|--------|
| Factor_1 | -4.75% | -1.89% | -0.68% | 0.69% | 1.10% | 5.36% | -0.87% | -1.05% |

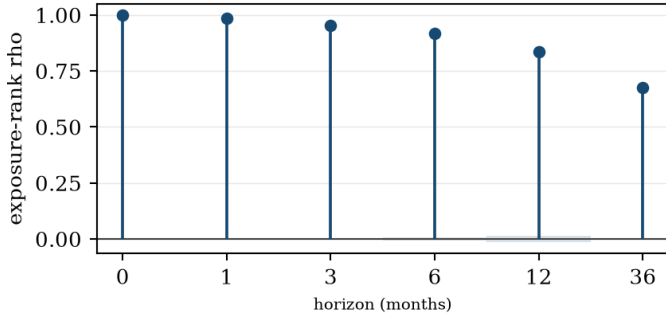


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

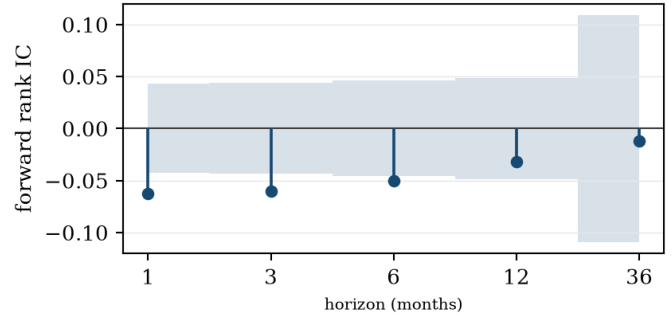


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direction                     | +1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Common months                 | 41                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Signals                       | Cluster_1(-1), Cluster_10(+1), Cluster_11(-1), Cluster_14(-1), Cluster_15(+1), Cluster_16(-1), Cluster_18(-1), Cluster_19(-1), Cluster_2(-1), Cluster_20(-1), Cluster_21(-1), Cluster_22(-1), Cluster_23(-1), Cluster_25(-1), Cluster_29(-1), Cluster_3(-1), Cluster_32(-1), Cluster_36(-1), Cluster_38(+1), Cluster_4(-1), Cluster_40(+1), Cluster_41(-1), Cluster_44(-1), Cluster_45(+1), Cluster_48(-1), Cluster_49(-1), Cluster_5(-1), Cluster_50(-1), Cluster_58(+1), Cluster_59(-1), Cluster_61(-1), Cluster_63(-1), Cluster_64(-1), Cluster_66(+1), Cluster_69(-1), Cluster_76(-1), Cluster_77(-1), Cluster_79(+1), Cluster_81(+1), Cluster_82(+1), Cluster_83(-1), Cluster_9(-1) |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Components required           | 26 of 42                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Available / comparison months | 43 / 41                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Available window              | 2023-01-01 to 2026-07-01                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Dropped: low breadth / common | 0 / 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Stocks, min / median / max    | 2093 / 2500 / 2723                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Available-history Sharpe      | -0.40                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Bootstrap Sharpe              | -0.10 [-0.96, 0.75]; Pr(S<=0)=61.3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Bootstrap tails               | skew=-0.08 [-0.54, 0.39]; max DD=-25.5% [-33.0%, -11.1%]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 4. Factor\_4

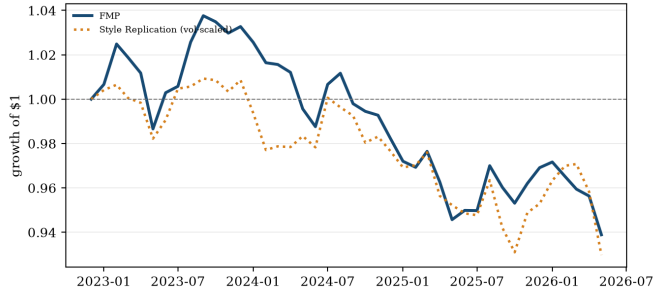


Figure: Common-window cumulative growth of \$1.

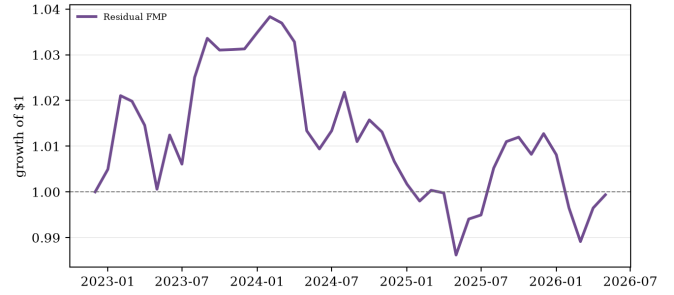


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|--------|----------|-------------|------------|
| Factor_4 | -1.77%    | 3.83%     | -0.46  | -0.94 | -9.51% | 0.480    | -0.35%      | 3.83%      |

| Factor   | MKT            | SMB            | VAL            | MOM              | BAB            | QMJ            | Alpha           | R2    |
|----------|----------------|----------------|----------------|------------------|----------------|----------------|-----------------|-------|
| Factor_4 | 0.08<br>(1.39) | 0.15<br>(3.60) | 0.10<br>(1.87) | -0.02<br>(-0.64) | 0.02<br>(0.38) | 0.18<br>(2.59) | 0.02%<br>(0.01) | 0.484 |

| Factor   | MKT   | SMB    | VAL    | MOM    | BAB    | QMJ    | Alpha | Total  |
|----------|-------|--------|--------|--------|--------|--------|-------|--------|
| Factor_4 | 1.72% | -0.77% | -0.49% | -0.23% | -0.06% | -1.96% | 0.02% | -1.77% |

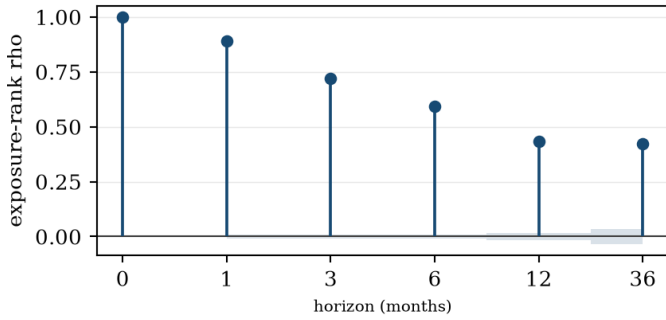


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

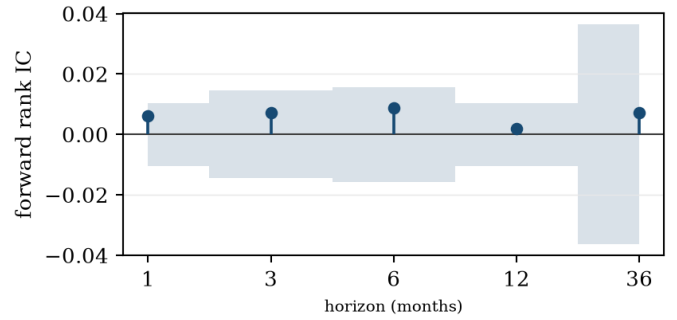


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| Direction                     | +1                                                                                                             |
| Common months                 | 41                                                                                                             |
| Signals                       | Cluster_30(+1), Cluster_31(+1), Cluster_42(+1), Cluster_46(-1), Cluster_54(-1), Cluster_62(-1), Cluster_65(+1) |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp                              |
| Components required           | 5 of 7                                                                                                         |
| Available / comparison months | 43 / 41                                                                                                        |
| Available window              | 2023-01-01 to 2026-07-01                                                                                       |
| Dropped: low breadth / common | 0 / 2                                                                                                          |
| Stocks, min / median / max    | 2044 / 2424 / 2591                                                                                             |
| Available-history Sharpe      | -0.08                                                                                                          |
| Bootstrap Sharpe              | -0.46 [-1.29, 0.20]; Pr(S<=0)=91.8%                                                                            |
| Bootstrap tails               | skew=0.34 [-0.09, 0.86]; max DD=-9.5% [-15.0%, -4.7%]                                                          |

## 5. Factor\_5

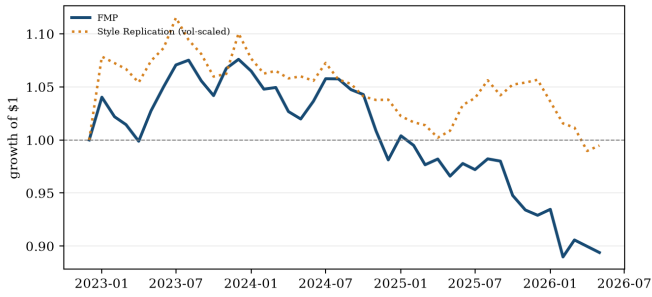


Figure: Common-window cumulative growth of \$1.

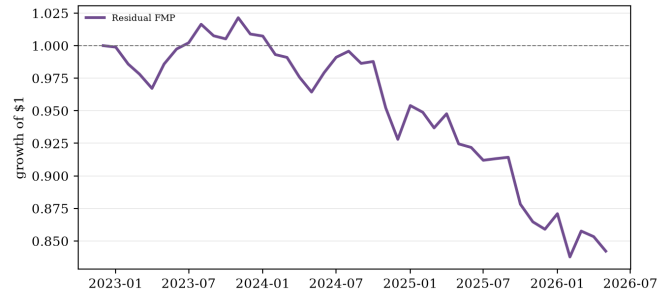


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

| Factor   | Ann. ret. | Ann. vol. | Sharpe | HAC t | Max DD  | Style R2 | Style alpha | Style vol. |
|----------|-----------|-----------|--------|-------|---------|----------|-------------|------------|
| Factor_5 | -3.07%    | 6.47%     | -0.47  | -1.01 | -17.31% | 0.261    | -3.10%      | 6.47%      |

| Factor   | MKT            | SMB            | VAL            | MOM              | BAB              | QMJ              | Alpha             | R2    |
|----------|----------------|----------------|----------------|------------------|------------------|------------------|-------------------|-------|
| Factor_5 | 0.16<br>(1.82) | 0.07<br>(0.60) | 0.02<br>(0.14) | -0.17<br>(-3.95) | -0.12<br>(-0.53) | -0.01<br>(-0.09) | -4.87%<br>(-1.38) | 0.284 |

| Factor   | MKT   | SMB    | VAL    | MOM    | BAB   | QMJ   | Alpha  | Total  |
|----------|-------|--------|--------|--------|-------|-------|--------|--------|
| Factor_5 | 3.52% | -0.34% | -0.08% | -1.73% | 0.30% | 0.12% | -4.87% | -3.07% |

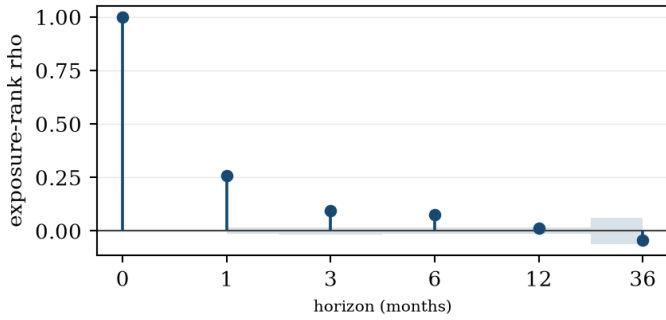


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

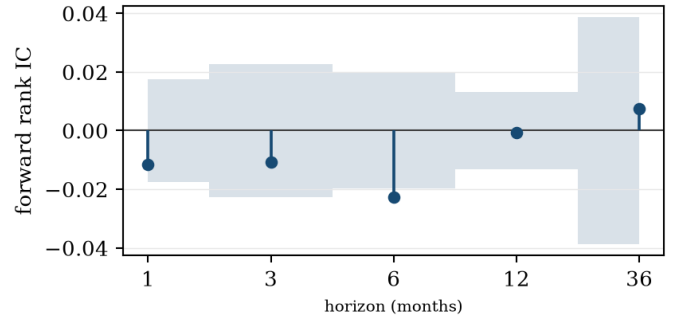


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

| Item                          | Value                                                                             |
|-------------------------------|-----------------------------------------------------------------------------------|
| Direction                     | +1                                                                                |
| Common months                 | 41                                                                                |
| Signals                       | Cluster_33(-1), Cluster_52(+1), Cluster_7(-1), Cluster_71(-1)                     |
| Effective construction        | rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp |
| Components required           | 3 of 4                                                                            |
| Available / comparison months | 43 / 41                                                                           |
| Available window              | 2023-01-01 to 2026-07-01                                                          |
| Dropped: low breadth / common | 0 / 2                                                                             |
| Stocks, min / median / max    | 2106 / 2547 / 2811                                                                |
| Available-history Sharpe      | -0.17                                                                             |
| Bootstrap Sharpe              | -0.47 [-1.28, 0.34]; Pr(S<=0)=87.9%                                               |
| Bootstrap tails               | skew=0.05 [-0.39, 0.51]; max DD=-17.3% [-25.3%, -6.1%]                            |



# Reproducibility

Run ID: 20260906T021608Z\_frozen-grouped-factor-model-oos-2023\_afa49d51c1.

Gold snapshot: version=20260829T051952Z.

Configuration hash: afa49d51c1.

Symbols requested / loaded: 0 / 3595.

Seed: 0. Comparison months: 41, 2023-01-01 to 2026-05-01.

Command:

C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/automatic\_factor\_grouping/handoff.py--source-rundata/outputs/automatic\_factor\_grouping/cluster\_only/20260906T021311Z\_cf908c3399.

Direction is frozen upstream; this report performs no return-based sign selection. The reported direction is applied consistently before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.